

Exam for Zeyang (Arthur) Yu, “The Mathematical Foundations of Political Methodology”

You have 3 hours for this exam. You may use a calculator and refer to your two class books (Pemberton & Rau and Bertsekas & Tsitsiklis). Please do not use the internet during the exam. Show your work: solutions without work may receive no credit.

1. Given the following hold, what (if anything) do we know about the relationship between sets A and C ?

- (a) $A \cap B = A$ and $B \cap C \neq \emptyset$
- (b) $A \cup B = A$ and $B \cap C \neq \emptyset$
- (c) $C \cup D = B$ and $B \cap C = A$

2. Find the limit of each of the following sequences as $n \rightarrow \infty$, if the limit exists.

(a)

$$\frac{4n^3 - 1}{(n + 1)^3}$$

(b)

$$\frac{(-1)^n n^{\frac{5}{8}}}{7 + n^{\frac{7}{8}}}$$

(c)

$$\frac{(-1)^n n^{\frac{1}{2}}}{n^{\frac{1}{2}} + n^{\frac{2}{5}}}$$

3. Use the (δ, ϵ) definition of a limit to prove that the function

$$f(x) = 1/2x - 10$$

is everywhere continuous.

4. Let

$$f(x) = x^2/(x + 4)$$

- (a) Find the critical points of f and characterize them as maxima, minima, or inflection points.
- (b) Find the regions on which f is convex and concave.
- (c) Find any asymptotes that f may have.

5. Perform the following matrix operations:

(a) Find the inverse of

$$\begin{bmatrix} 1 & 1 & 4 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

(b) Find the rank of

$$\begin{bmatrix} 1 & 1 & 5 \\ 0 & 2 & 4 \\ -2 & 0 & -6 \\ 1 & 9 & 21 \end{bmatrix}$$

(c) Check whether the following matrix is positive definite, positive semidefinite, negative definite, negative semidefinite, or none of the above.

$$\begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix}$$

6. Find and classify all of the critical points of the following functions:

(a)

$$f(x, y) = -10x^2 + 4xy - \frac{2y^2}{5}$$

(b)

$$f(x, y, z) = -x^2 + \log(x) - \frac{1}{y} - y - z^2 + z \quad \text{for } x, y, z > 0$$

(c)

$$f(x, y) = x^2 + y^2 - xy \text{ subject to the constraint that } 2x - y = 3$$